

Haipeng Gan

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EDUCATION

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- **Nanjing University** Nanjing, China
Master of Science in Chemistry Sept. 2024 - June 2027 (expected)
 - **Nanjing University** Nanjing, China
Bachelor of Science in Computer Science and Technology Sept. 2020 - June 2024

RESEARCH INTERESTS

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- **Summary:** My research has evolved from model-guided peptide inhibitor screening, to scalable reaction-language-model training, to AI-assisted discovery of catalytic self-assembling peptides. Across these projects, I am interested in building AI-for-science workflows that turn sparse literature, molecular sequences, and experimental data into testable candidates for chemical and biomolecular discovery.

RESEARCH EXPERIENCE

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- **Dong Lab, Nanjing University** Nanjing, China
Graduate Researcher Oct. 2023 – Present
 - **Starting Point: Peptide Inhibitor Design:** My first research project used machine learning to support Hsp90 peptide inhibitor screening. I learned how model scores, docking results, and sequence mutations can be combined to refine peptide candidates, and I began to see molecular design as a ranking and decision-making problem rather than only a prediction task.
 - **Decision-Making Under Imperfect Evidence:** The JCIM project also exposed a recurring problem in molecular discovery: different evidence sources do not always agree. Prediction scores, docking heuristics, and sequence mutations each provide partial information, so candidate selection requires judging which proxy signal is reliable enough for the next step.
 - **From Prediction To Chemical Language Models:** After the JCIM peptide project, I joined ChemBART, a reaction-language-model project for synthesis planning. This shifted my focus from small supervised models to chemical foundation models, and gave me experience with model deployment, profiling, pretraining bottlenecks, and scalable training pipelines.
 - **Engineering For Scientific Models:** ChemBART taught me that AI-for-science models depend heavily on data and systems engineering. I improved training throughput by removing repeated tokenization, grouping sequences by length, reducing unnecessary precision, and using distributed acceleration, which made large-scale reaction-model pretraining more practical.
 - **Tool Reliability Perspective:** Working on a reaction-language model also made me think about scientific AI systems as collections of imperfect tools. A synthesis model, a docking engine, or a predictor is only useful when its uncertainty, failure modes, and computational cost are considered during decision-making.
 - **Toward Discovery Loops: SAPIENS:** My most complete project is SAPIENS, where I connected LLM-assisted literature mining, dataset construction, positive-unlabeled learning, multi-task modeling, and candidate prioritization for catalytic self-assembling peptide discovery. This project turned my earlier modeling and engineering experience into a full discovery workflow.
 - **Learning From Biased Literature:** SAPIENS made the data problem explicit: peptide self-assembly literature contains many positive reports but few reliable negatives. I learned to treat literature mining as evidence construction, using positive-unlabeled learning and careful curation to build a dataset that could support downstream discovery decisions.
 - **Learning From Experimental Validation:** In SAPIENS, experimental co-first authors performed the wet-lab validation, while I led the computational pipeline and much of the analysis and writing. Working across this boundary taught me that useful AI models must produce candidates that are interpretable, chemically plausible, and realistic for collaborators to test.
 - **Current Research Direction:** The sequence from JCIM to ChemBART to SAPIENS shaped my PhD direction: I want to build evidence-grounded AI discovery systems that connect literature, curated datasets, molecular representation learning, uncertainty-aware evaluation, and decision-making under sparse, delayed experimental feedback.

SELECTED PUBLICATIONS

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- **Gan, H.#, Wang, X.#, Liang, G.#, Zhang, F., Xie, J., and Dong, H.*:** *AI Unlocks Superior Catalytic Self-Assembly in Peptides*. Submitted to *JACS* after substantial revision informed by external peer review at *Nature Catalysis*; ChemRxiv preprint available at chemrxiv.org/doi/full/10.26434/chemrxiv.15002365/v1. First-listed co-first author; led LLM data mining, dataset construction, machine learning model development, computational analysis, and major manuscript writing.
 - **Zang, M.#, Gan, H.#, Zhou, X.#, Wang, L., and Dong, H.*:** *Dual-site targeting by peptide inhibitors of the N-Terminal domain of Hsp90: mechanism and design*. *Journal of Chemical Information and Modeling* 65(10), 5113–5123 (2025). Co-first author; designed the two-mutation screening strategy, deployed the machine-learning model, ran inference, and analyzed correlations with docking scores. [doi:10.1021/acs.jcim.5c00629](https://doi.org/10.1021/acs.jcim.5c00629).
 - **Li, K.#, Zhang, Y.#, Wang, J.#, Gan, H., Sun, Z., Lei, X., and Dong, H.*:** *ChemBART: Pre-training a Reaction Language Model for Comprehensive Synthesis Planning*. Under review at *Nature Communications*. Contributed to model deployment, profiling, and pretraining acceleration for a reaction-language-model framework.

RESEARCH CONTRIBUTIONS

- **AI Unlocks Superior Catalytic Self-Assembly in Peptides:** Submitted to *JACS* after revision informed by external peer review at *Nature Catalysis*; ChemRxiv preprint available.
 - First-listed co-first author.
 - Built the LLM-assisted literature-mining pipeline over 5,606 peptide self-assembly papers.
 - Curated 3,961 positive self-assembling peptide sequences and reconstructed 26,654 high-confidence negative examples.
 - Trained Transformer-based multi-task models to transfer self-assembly knowledge to sparse catalytic-function prediction.
 - Screened a candidate space of over 9.4 million HXH-motif heptapeptide variants and helped prioritize experimentally testable candidates.
 - Learned that AI-driven discovery is often limited by sparse and delayed supervision, motivating my interest in closed-loop discovery systems and uncertainty-aware candidate prioritization.
 - Helped analyze why selected sequences deviated from conventional alternating hydrophobic/hydrophilic design rules.
 - Contributed substantially to computational analysis, figure preparation, interpretation, and manuscript writing.
- **Dual-site targeting by peptide inhibitors of the N-Terminal domain of Hsp90: mechanism and design:** *Journal of Chemical Information and Modeling* 65(10), 5113–5123 (2025).
 - Co-first author.
 - Designed the two-mutation screening strategy for peptide inhibitor refinement.
 - Deployed the trained machine-learning model and ran inference for candidate prioritization.
 - Analyzed the relationship between prediction scores and docking results to assess whether the model ranking was chemically consistent.
 - Focused on a peptide-targeting setting where sequence optimization, binding prediction, and docking-based heuristics had to be considered together.
 - Helped connect model-guided screening to a practical dual-site targeting mechanism for the Hsp90 N-terminal domain.
- **ChemBART: Pre-training a Reaction Language Model for Comprehensive Synthesis Planning:** *Nature Communications* manuscript under review.
 - Contributed to model deployment and pretraining optimization.
 - Profiled the pretraining workflow and identified repeated tokenization, sequence length imbalance, and unnecessary numerical precision as major bottlenecks.
 - Improved the pipeline with tokenization caching, sequence-length binning, precision reduction, and ColossalAI-based distributed acceleration.
 - Achieved about a 56x speedup relative to an inefficient earlier implementation, making reaction-language-model pretraining practical on the local cluster.
 - Gained experience with reaction-model pretraining and adaptation for synthesis-planning tasks.

TECHNICAL SKILLS

- **Programming:** Python; basic C/C++.
- **Machine Learning:** PyTorch, Transformer models, multi-task learning, positive-unlabeled learning, active learning, model evaluation.
- **LLM and Data Mining:** Literature mining, information extraction, prompt design, dataset construction, data cleaning, automated evaluation.
- **Scientific Computing:** Linux, Git, Python data stack, concurrency, profiling, HPC workflows.
- **Chemistry and Modeling:** Peptide self-assembly, catalytic peptide design, cheminformatics, docking-score analysis, molecular visualization.

ENGLISH PROFICIENCY

TOEFL iBT (new format), taken in May 2026: Total 6/6; Reading 6/6; Listening 6/6; Writing 5.5/6; Speaking 5.5/6.

AWARDS AND HONORS

- First-Class Academic Scholarship, Nanjing University; ranked 1/12 in the mid-term postgraduate assessment.
- Huatai Securities Technology Scholarship, Nanjing University.

TEACHING AND COLLABORATION

- **Teaching Assistant, Calculus Problem Sessions** Nanjing University
Led three semesters of problem sessions Feb. 2025 – June 2026
- Led undergraduate calculus problem sessions for three semesters and explained problem-solving strategies clearly and efficiently.
- Helped lab colleagues troubleshoot Python, machine-learning, and presentation issues.
- Developed a habit of explaining technical ideas in a structured way, which also improved my manuscript writing and group communication.

REFERENCES

Available upon request. Primary referee: Prof. Hao Dong, Nanjing University.